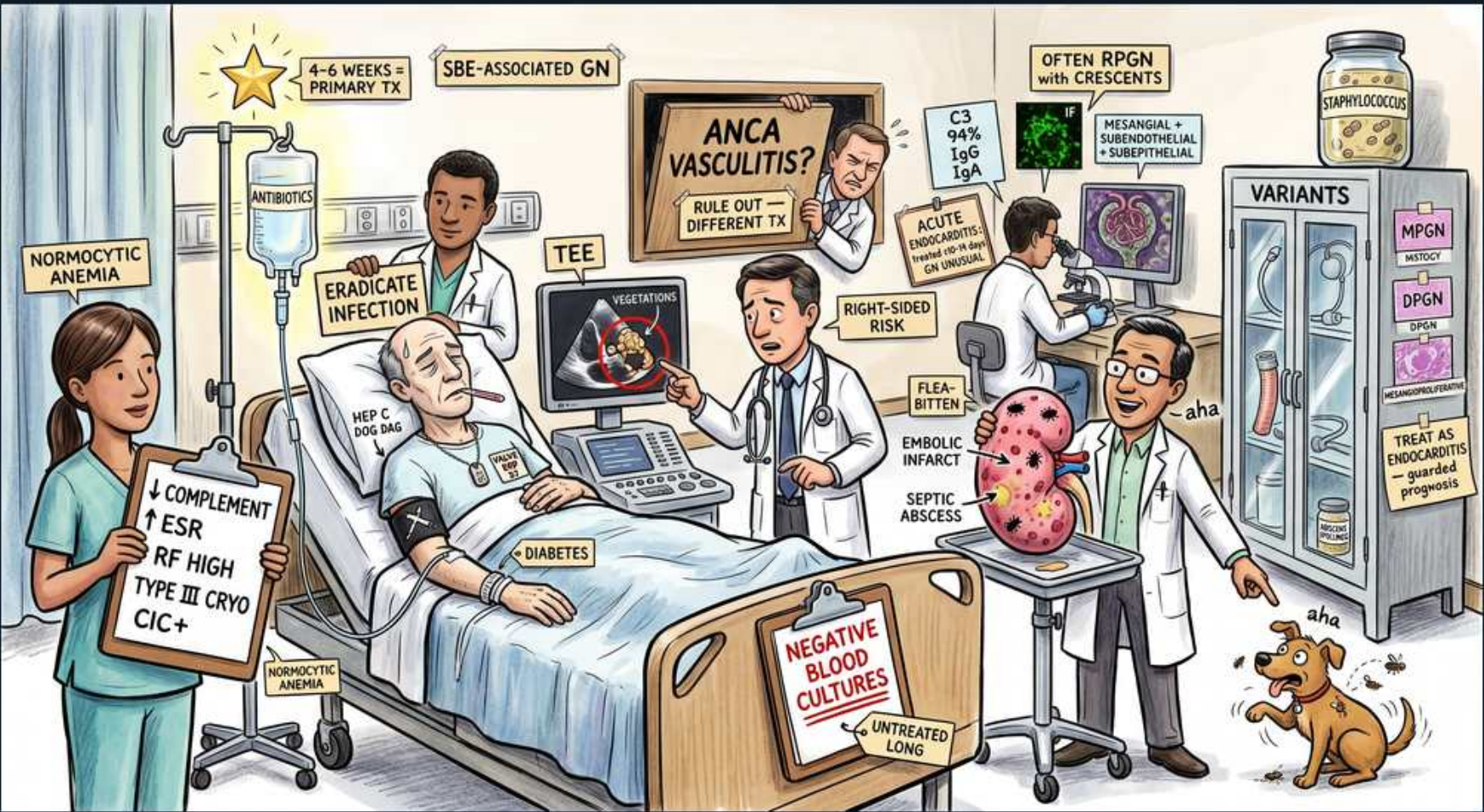


Endocarditis-Associated (SBE) Glomerulonephritis



1. SBE-associated GN

WHAT YOU SEE

Banner 'SBE-associated GN', center top

WHAT IT ENCODES

Subacute bacterial endocarditis can cause immune-complex glomerulonephritis from chronic bacteremia. Presents with hematuria, proteinuria, AKI, and low complement during active infection. Part of the broader infection-related GN family.

BOARD TRAP

Endocarditis GN is immune-complex mediated and complement-consuming — it can mimic both lupus and ANCA vasculitis, the two classic mimics tested.

2. Eradicate the infection = primary Tx

WHAT YOU SEE

'Eradicate infection' / antibiotics, left

WHAT IT ENCODES

The cornerstone of treatment is ANTIBIOTICS to clear the infection (typically 4–6 weeks); renal function and complement usually recover as bacteremia is cleared. Surgery if valve indication. Immunosuppression is generally avoided.

BOARD TRAP

Treating endocarditis GN with steroids/immunosuppression instead of antibiotics is the trap — you must eradicate the infection first.

3. 4–6 weeks primary Tx

WHAT YOU SEE

Gold star '4–6 weeks primary Tx', upper left

WHAT IT ENCODES

Prolonged antibiotic course (often 4–6 weeks of IV therapy) is required to sterilize valvular vegetations and resolve the immune-complex GN. Monitor complement normalization as a marker of response.

BOARD TRAP

A short antibiotic course risks relapse — endocarditis demands a full prolonged course, and the GN tracks infection control.

4. ANCA vasculitis mimic

WHAT YOU SEE

'ANCA vasculitis? rule out — different Tx' sign, center top

WHAT IT ENCODES

Endocarditis can produce POSITIVE ANCA (often PR3/c-ANCA) and a pauci-immune-appearing crescentic GN — a dangerous mimic. The treatments diverge: antibiotics for endocarditis vs immunosuppression for primary ANCA vasculitis.

BOARD TRAP

A positive ANCA does NOT equal primary vasculitis — blood cultures and echo must exclude endocarditis before immunosuppressing, or you'll seed infection.

5. Low complement / high ESR / RF

WHAT YOU SEE

Nurse card: ↓ complement, ESR high, type III cryo, CIC+

WHAT IT ENCODES

Lab signature of active endocarditis GN: LOW C3/C4 (complement consumption), high ESR/CRP, positive rheumatoid factor, circulating immune complexes, and type III cryoglobulins. Complement normalizes with infection control.

BOARD TRAP

Low complement + positive RF + AKI can look like lupus or cryoglobulinemic MPGN — the febrile patient with a murmur and positive cultures points to endocarditis.

6. Normocytic anemia

WHAT YOU SEE

'Normocytic anemia' label, left

WHAT IT ENCODES

Anemia of chronic disease (normocytic, normochromic) accompanies subacute endocarditis from sustained inflammation. Other systemic signs: fever, splenomegaly, embolic phenomena, and constitutional symptoms.

BOARD TRAP

Normocytic anemia + fever + new murmur should trigger blood cultures and echo — attributing it to the renal failure alone misses the infection.

7. TEE / vegetations

WHAT YOU SEE

TEE screen showing vegetations, center

WHAT IT ENCODES

Transesophageal echo is more sensitive than TTE for valvular vegetations and is key to diagnosis when SBE-GN is suspected. Combined with blood cultures it fulfills the Duke criteria.

BOARD TRAP

A negative transthoracic echo does NOT exclude endocarditis — proceed to TEE if suspicion is high (e.g., GN with positive cultures).

8. Negative blood cultures pitfall

WHAT YOU SEE

Red sign 'negative blood cultures — untreated long'

WHAT IT ENCODES

Culture-negative endocarditis (prior antibiotics, fastidious HACEK or Coxiella/Bartonella organisms) delays diagnosis. Sustained untreated infection drives ongoing GN and worse renal outcomes.

BOARD TRAP

Negative cultures don't rule out endocarditis — serologies for Bartonella/Coxiella and prolonged incubation are needed; missing it leaves the GN untreated.

9. Right-sided risk (IVDU)

WHAT YOU SEE

'Right-sided risk' label, center

WHAT IT ENCODES

Right-sided endocarditis (tricuspid) is associated with IV drug use, often Staph aureus, with septic pulmonary emboli. Acute Staph endocarditis can cause a more aggressive, crescentic GN than classic subacute strep disease.

BOARD TRAP

Acute Staph endocarditis GN can be crescentic/RPGN and progress fast — don't assume all endocarditis GN is the indolent subacute strep type.

10. IF pattern: C3, IgM, IgG

WHAT YOU SEE

'C3 / IgM / IgG, mesangial+subendothelial+subepithelial' microscopy, center

WHAT IT ENCODES

Immunofluorescence shows granular C3 (often with IgM/IgG) in mesangial, subendothelial, and subepithelial locations — true immune-complex deposition. EM may show subepithelial humps resembling post-infectious GN.

BOARD TRAP

Hump-like subepithelial deposits make endocarditis GN resemble post-strep GN on EM — clinical setting (endocarditis vs antecedent pharyngitis) separates them.

11. Embolic infarct / septic abscess

WHAT YOU SEE

Flea-bitten kidney with infarcts/abscess, center right

WHAT IT ENCODES

Beyond immune-complex GN, septic emboli from vegetations can cause renal INFARCTS and abscesses ('flea-bitten kidney'), and acute interstitial nephritis from antibiotics — multiple mechanisms of renal injury coexist.

BOARD TRAP

Renal injury in endocarditis isn't only glomerular — embolic infarcts, abscess, and drug-induced AIN also contribute and change management.

12. Variants: MPGN / DPGN crescents

WHAT YOU SEE

Variants cabinet: MPGN, DPGN, treat as endocarditis

WHAT IT ENCODES

Histologic patterns vary: diffuse proliferative GN, focal proliferative GN, MPGN-like, or crescentic GN. Regardless of pattern, the treatment is to ERADICATE the infection rather than immunosuppress.

BOARD TRAP

Whatever the biopsy pattern, the answer remains antibiotics — the histologic label doesn't override 'treat the underlying infection.'
